Northern University of Business and Technology Khulna

Department of Computer Science and Engineering

Course Title: Software Development III

Course Code: CSE 4104

Team Name: CSE4104-7B-T05

Section: 7B

Project Title:

AI Inventory and Sales Forecasting System

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GitHub Repository Link:

https://github.com/nishansami32-ux/CSE-4104-7B-T05-AI-Inventory-and-Sales-Forecasting-System.git

Submitted to:

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Project Title:

AI Inventory and Sales Forecasting System

Project Overview :

The AI Inventory and Sales Forecasting System is a web-based system that helps small shops manage products, monitor sales, predict demand and identify items likely to run out of stock. The system will combine everyday inventory operations with AI-supported demand forecasting. The system will analyze past sales data to predict future product demand and provide low-stock alerts and reorder suggestions. This will help shop owners make better purchasing decisions.

The proposed technology stack is React.js for the frontend, Node.js with Express.js for the backend, PostgreSQL for data storage, and an AI service using OpenAI and/or Gemini where appropriate. Vercel or Render is proposed for deployment.

Problem Statement:

Existing Problems

Many small shops still manage their stock with notebooks or guesswork. Owners often discover that an item is finished only when a customer asks for it, which causes lost sales and unhappy customers. At the same time, they may buy too much of slow-moving products, which ties up their limited money and can lead to waste or expiry. Manual record keeping is also slow, error-prone and gives no clear idea of which products sell best.

Importance of the Project

- Overstocking: Increases storage costs and blocks business capital.
- Stock Shortages: Cause missed sales and customer dissatisfaction.
- Manual Record-Keeping: Takes time and increases data entry errors.
- Inaccurate Inventory Records: Duplicate entries and delayed updates create incorrect stock information.
- Poor Sales Analysis: Makes it difficult to understand sales patterns.
- Product Identification: Difficult to identify fast-moving, slow-moving, and low-stock products.
- Poor Purchasing Decisions: Decisions based on guesswork can lead to overstocking or stockouts.
- Lack of Demand Forecasting: Traditional systems cannot accurately predict future product demand.
- Increased Business Costs: Excess inventory increases storage and holding costs.
- Profit Loss: Stock shortages and excess inventory reduce business profits.

Limitations of Existing Solutions

Existing tools like Zoho Inventory, inFlow, and Odoo provide inventory management, but they can be costly and complex for small businesses. Their demand forecasting and intelligent recommendations may also be limited. Our proposed system aims to provide an affordable, user-friendly platform with AI-based sales forecasting, low-stock alerts, and smart reorder recommendations.

Objectives

- To develop a web-based system that manages products, suppliers, sales, and purchases efficiently.
- To analyze historical sales data and predict future product demand using AI and machine learning techniques.
- To reduce overstocking and prevent stock shortages by monitoring inventory levels.
- To suggest appropriate reorder quantities and identify the right time to purchase new products.
- To identify sales trends, seasonal patterns, and slow-moving products through data analysis.
- To integrate an AI assistant that explains sales forecasts, provides intelligent recommendations, and answers business-related queries.
- To provide a dashboard with stock updates, low-stock alerts, sales reports, and visual charts.
- To help small and medium-sized businesses reduce operational costs, save time, and make better inventory and purchasing decisions.
- To implement a secure login system with appropriate access for business owners and staff.

Features

- **Authentication:** Secure registration and login with Owner and Staff roles.
- **Product Management:** Add, edit, delete and search products with category, price, stock level and supplier.
- **Sales and Purchase Recording:** Record sales and incoming stock; stock updates automatically.
- **Dashboard:** Inventory value, low-stock alerts, pending orders and sales trend charts.
- **Demand Forecasting:** Predicts the next 7 to 30 days of demand for each product.
- **Stock-out Prediction:** Estimates days of stock left and flags high-risk products.
- **Reorder Recommendation:** Suggests how much to order based on forecast, current stock and supplier lead time.
- **AI Assistant:** Explains forecasts and recommendations in simple language and answers questions such as "What should I reorder this week?".
- **Reports:** Daily and monthly sales, best-selling and slow-moving products.

System Workflow

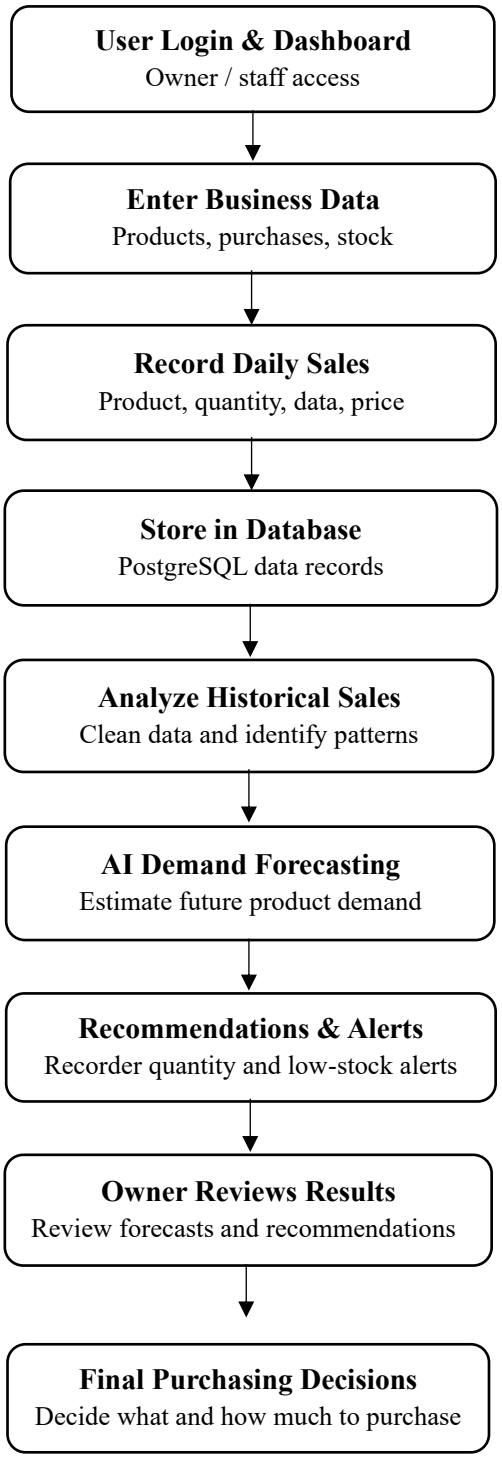


Figure 1: Workflow diagram of the AI Inventory and Sales Forecasting System

Technology Stack

Frontend:

- React.js

Backend:

- Node.js + Express.js

Database:

- PostgreSQL

AI Integration:

- OpenAI API / Gemini API

Deployment:

- Vercel / Render

Expected Outcome

Expected User Benefits

- Manage products, suppliers, purchases, sales, and stock in one platform.
- Get sales reports and future demand forecasts.
- Receive low-stock alerts and smart reorder recommendations.
- Reduce stock-outs and lost sales.
- Avoid overstocking and unnecessary costs.
- Get simple advice on what and how much to reorder.
- Save time and reduce manual work.
- Make better business decisions using sales trends.

The System Improves the Existing Workflow

The shop moves from manual notebooks and guesswork to a digital, data-driven process. Instead of reacting after an item runs out, the owner is warned in advance and receives a recommended order quantity. Routine counting and calculation are automated, saving time and reducing errors.

Expected benefits of AI integration

AI will analyze historical sales data to predict future product demand. It will identify sales patterns, predict potential stock shortages, and recommend suitable reorder quantities. This will help businesses reduce overstocking, prevent stock shortages, minimize financial losses, and improve inventory planning.